



BAHIR DAR UNIVERSITY

BAHIRDAR INSTITUTE OF TECHNOLOGY(BiT)

FACULTY OF COMPUTING

DEPARTMENT OF INFORMATION TECHNOLOGY (ITBED)

OPERATING SYSTEM PROJECT WORK

Operating System: OmniOS

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Name

ID Number

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Submitted to:Lec.Wendimu

1. Introduction

- An operating system is the core software that manages computer hardware and provides services for application programs. Modern computing environments increasingly rely on stable, secure, and efficient operating systems, especially in server and enterprise contexts.
- OmniOS is a Unix-like operating system based on the illumos open-source operating system, which originates from Oracle Solaris. OmniOS is designed mainly for servers, storage systems, and virtualization environments. It is well known for its stability, performance, and advanced features such as the Zettabyte File System (ZFS).
- The motivation for this project is to gain practical experience in installing and working with a non-Linux, enterprise-grade operating system in a virtual environment. Using OmniOS helps students understand advanced operating system concepts such as boot processes, file systems, memory management, and virtualization without requiring physical hardware.

2. Objectives of the Project

The objectives of this project work are:

- To understand the process of installing an operating system in a virtual environment.
- To apply theoretical operating system concepts in a practical setup.
- To gain hands-on experience with OmniOS and its features.
- To understand the role of ZFS in modern operating systems.
- To develop troubleshooting skills during OS installation and configuration.
- To understand the importance of virtualization in modern computing.

3. Requirements

3.1 Hardware Requirements

- 64-bit processor (Intel or AMD)
- Minimum 4 GB RAM (8 GB recommended)
- At least 30 GB free disk space
- Internet connection (optional but recommended)

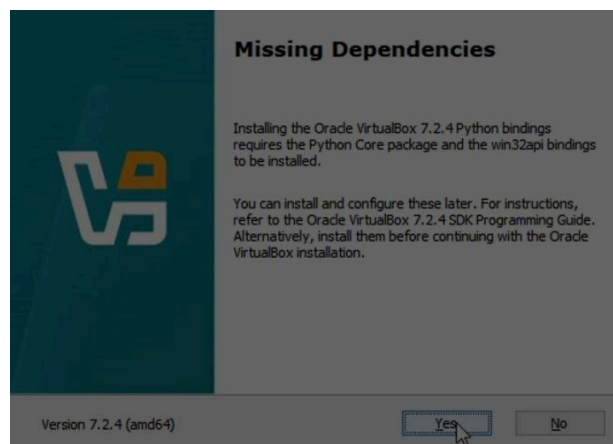
3.2 Software Requirements

- OmniOS ISO image
- Oracle VM VirtualBox or VMware Workstation
- Host operating system (Windows, Linux, or macOS)

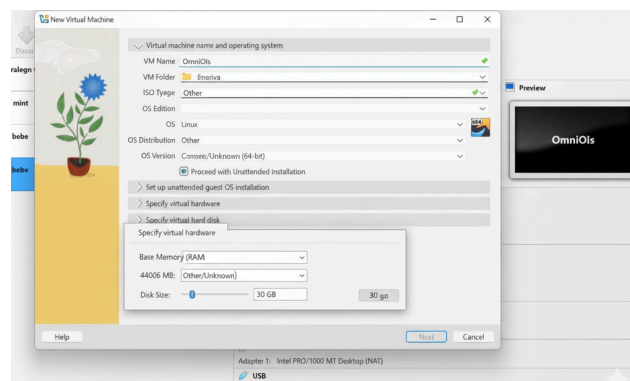
4. Installation Steps of OmniOS in a Virtual Environment

4.1 Creating the Virtual Machine

- install a virtual box

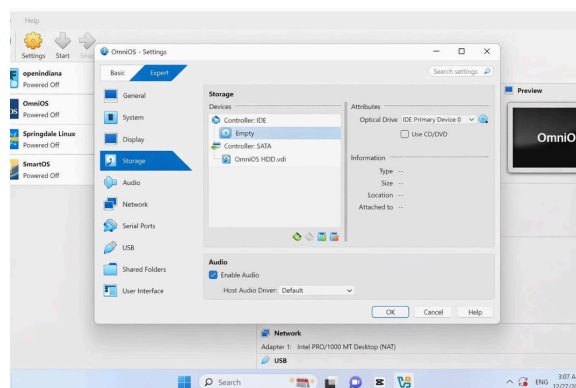


1. Open VirtualBox.
2. Click on **New** to create a new virtual machine.
3. Enter the VM name as **OmniOS**.
4. Select **Type: Other** and **Version: Other/Unknown (64-bit)**.
5. Allocate at least 4 GB of RAM.
6. Create a virtual hard disk with a minimum size of 30 GB.



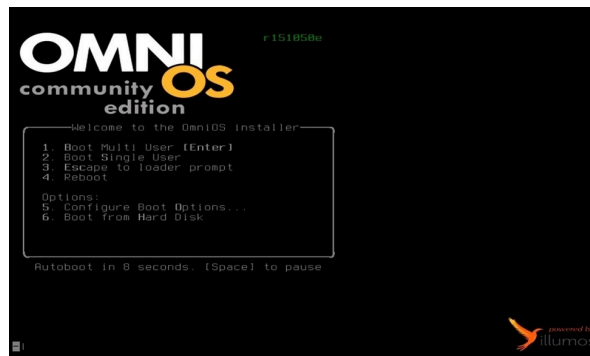
4.2 Attaching the OmniOS ISO

1. Go to VM **Settings**.
2. Select **Storage**.
3. Attach the OmniOS ISO file to the virtual optical drive.

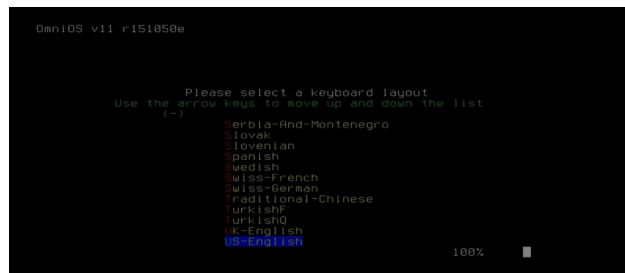


4.3 Installing OmniOS

1. Start the virtual machine.
2. Select the installation option from the boot menu.

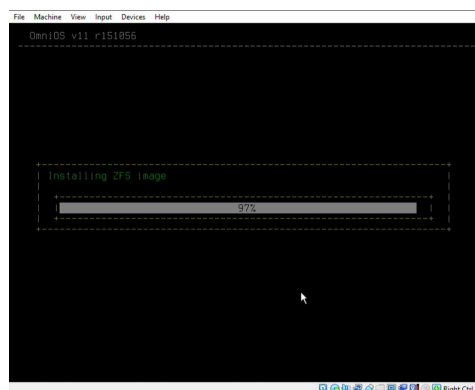


3. Choose language, keyboard layout, and time zone.



4. Configure network settings (automatic configuration recommended).

5. Select disk and create ZFS root pool (default settings).



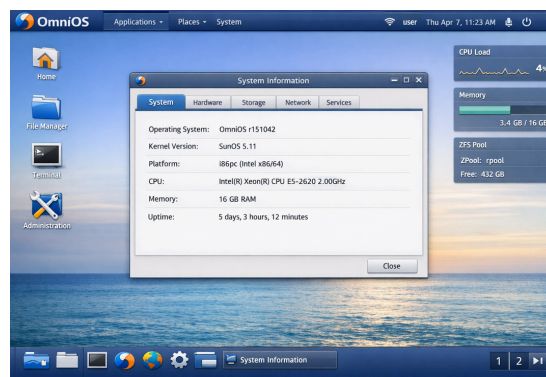
6. Create a user account using your **full name**.

7. Set root password.



8. Confirm installation settings and begin installation.

9. Reboot the system after installation completes.



5. Issues Faced During Installation

Some common issues faced during the installation process include:

- Network interface not detected automatically.
- Difficulty understanding ZFS pool configuration.
- Boot issues after installation.

6. Solutions to the Problems

- Changed network adapter to **Intel PRO/1000** in VirtualBox settings.
- Used default ZFS configuration recommended by the installer.
- Rechecked ISO file integrity and VM boot order.

7. File System Support

OmniOS mainly supports the **Zettabyte File System (ZFS)**.

Why ZFS?

- Provides data integrity through checksums.
- Supports snapshots and rollback features.
- High performance and scalability.
- Integrated volume management.

Other file systems are limited compared to ZFS, making ZFS the preferred and recommended file system for OmniOS.

8. Advantages and Disadvantages of OmniOS

Advantages

- High stability and reliability.
- Excellent ZFS support.
- Suitable for servers and virtualization.
- Strong security features.

Disadvantages

- Not user-friendly for beginners.
- Limited graphical user interface.
- Smaller community compared to Linux distributions.

9. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is the technology that allows multiple operating systems to run on a single physical machine by creating virtual machines.

Why Virtualization?

- Efficient use of hardware resources.
- Safe environment for testing operating systems.
- Cost-effective and flexible.

How Virtualization Works?

A hypervisor such as VirtualBox or VMware creates and manages virtual machines. OmniOS runs as a guest operating system on top of the host system.

10. Conclusion

In this project, OmniOS was successfully installed and configured in a virtual environment. The project provided practical knowledge of enterprise operating systems, ZFS file systems, and virtualization technology. It enhanced both theoretical understanding and practical skills in operating system administration.

11. Future Outlook and Recommendation

OmniOS is highly recommended for server environments, storage solutions, and virtualization platforms. Future work may include advanced ZFS administration, network services configuration, and performance optimization. Students are encouraged to further explore OmniOS for real-world system administration experience.

THE END